

## **Day 02 Assignment — Understanding My Network**

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### **1. What is a computer network?**

**A.** A computer network is a digital telecommunication which allows nodes to share resources.

### **2. What is the difference between a client and a server?**

**A.** Client: The device which accesses the services provided by the server  
Server : The device which produces the functions and services.

### **3. What is a LAN?**

**A.** LAN (Local Area Network) is a network that connects devices within a small geographical area, such as a home, office, school, or building.

### **4. What is a WAN?**

**A.** WAN (Wide Area Network) connects networks across a large geographical area, such as cities, countries, or continents.

### **5. What is the purpose of a switch?**

**A.** A switch connects multiple devices within a LAN and forwards data to the correct device using MAC addresses.

### **6. What is the purpose of a router?**

**A.** A router connects different networks and forwards packets between them using IP addresses.

### **7. What is a firewall?**

**A.** A firewall is a security device or software that controls network traffic based on predefined rules.

It can allow or block traffic based on things such as:

- IP address
- Port
- Protocol
- Application

## **8. What is an IP address?**

**A.** An IP address is a logical address assigned to a device on a network so that it can be identified and communicated with.

## **9. What is a MAC address?**

**A.** A MAC (Media Access Control) address is a hardware/network-interface address used to identify a device on a local network.

## **10. What is a default gateway?**

**A.** A default gateway is the device, usually a router, that a computer sends traffic to when the destination is outside its local network.

## **11. What is DNS?**

**A.** DNS (Domain Name System) translates domain names into IP addresses.

## **12. Why is networking knowledge important for cybersecurity?**

**A.** Networking knowledge is important in cybersecurity because most attacks involve network communication.

It helps security professionals understand IPs, ports, protocols, traffic, and network logs.

It is essential for detecting and investigating attacks such as scanning, malware communication, and data theft.

## **13. Write your computer's private IPv4 address.**

A. 192. 168. 204. 1

**14. Write your default gateway.**

A. fe80: :c69a:31ff:fe63:d4b0%17

**15. Write your DNS server.**

A. 192. 168. 1. 1

**Ipconfig:**

```
Command Prompt
Microsoft Windows [Version 10.0.22631.6199]
(c) Microsoft Corporation. All rights reserved.

C:\Users\deepa>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::f704:b4bf:4fee:7e3d%18
    IPv4 Address. . . . . : 192.168.204.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::5eb7:10bf:9844:b263%11
    IPv4 Address. . . . . : 192.168.149.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

```
Command Prompt

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2401:4900:882f:908f:93d3:9652:21dc:7c62
    Temporary IPv6 Address. . . . . : 2401:4900:882f:908f:942f:2a68:c851:9cda
    Link-local IPv6 Address . . . . . : fe80::bd26:fdc5:3e24:3bb7%17
    IPv4 Address. . . . . : 192.168.1.7
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::c69a:31ff:fe63:d4b0%17
                                192.168.1.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

## Ping:

```
C:\Users\deepa>ping google.com

Pinging google.com [2404:6800:4007:82f::200e] with 32 bytes of data:
Reply from 2404:6800:4007:82f::200e: time=31ms
Reply from 2404:6800:4007:82f::200e: time=35ms
Reply from 2404:6800:4007:82f::200e: time=25ms
Reply from 2404:6800:4007:82f::200e: time=30ms

Ping statistics for 2404:6800:4007:82f::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 25ms, Maximum = 35ms, Average = 30ms
```

## tracert:

```
C:\Users\deepa>tracert google.com

Tracing route to google.com [2404:6800:4007:82f::200e]
over a maximum of 30 hops:

  0  0 ms  0 ms  0 ms  2401:4900:882f:908f:c69a:31ff:fe63:d4b0
  1  *      *      *      Request timed out.
  2  35 ms  26 ms  26 ms  fc00::83
  3  32 ms  34 ms  49 ms  2404:a800:3a00:201::cd
  4  27 ms  36 ms  31 ms  2404:a800::92
  5  30 ms  38 ms  35 ms  pnmaaa-be-in-x0e.1e100.net [2404:6800:4007:82f::200e]

Trace complete.
```

